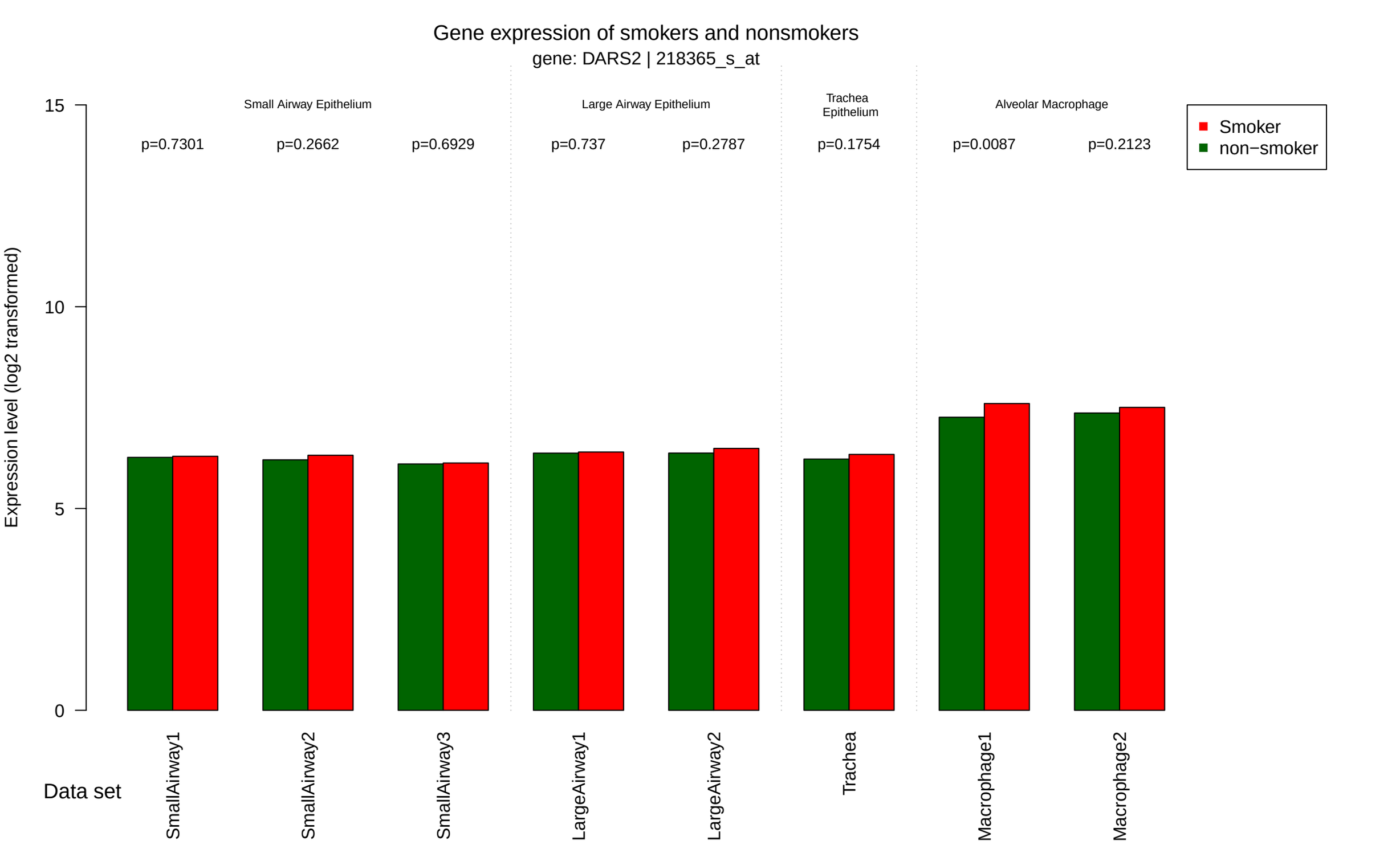
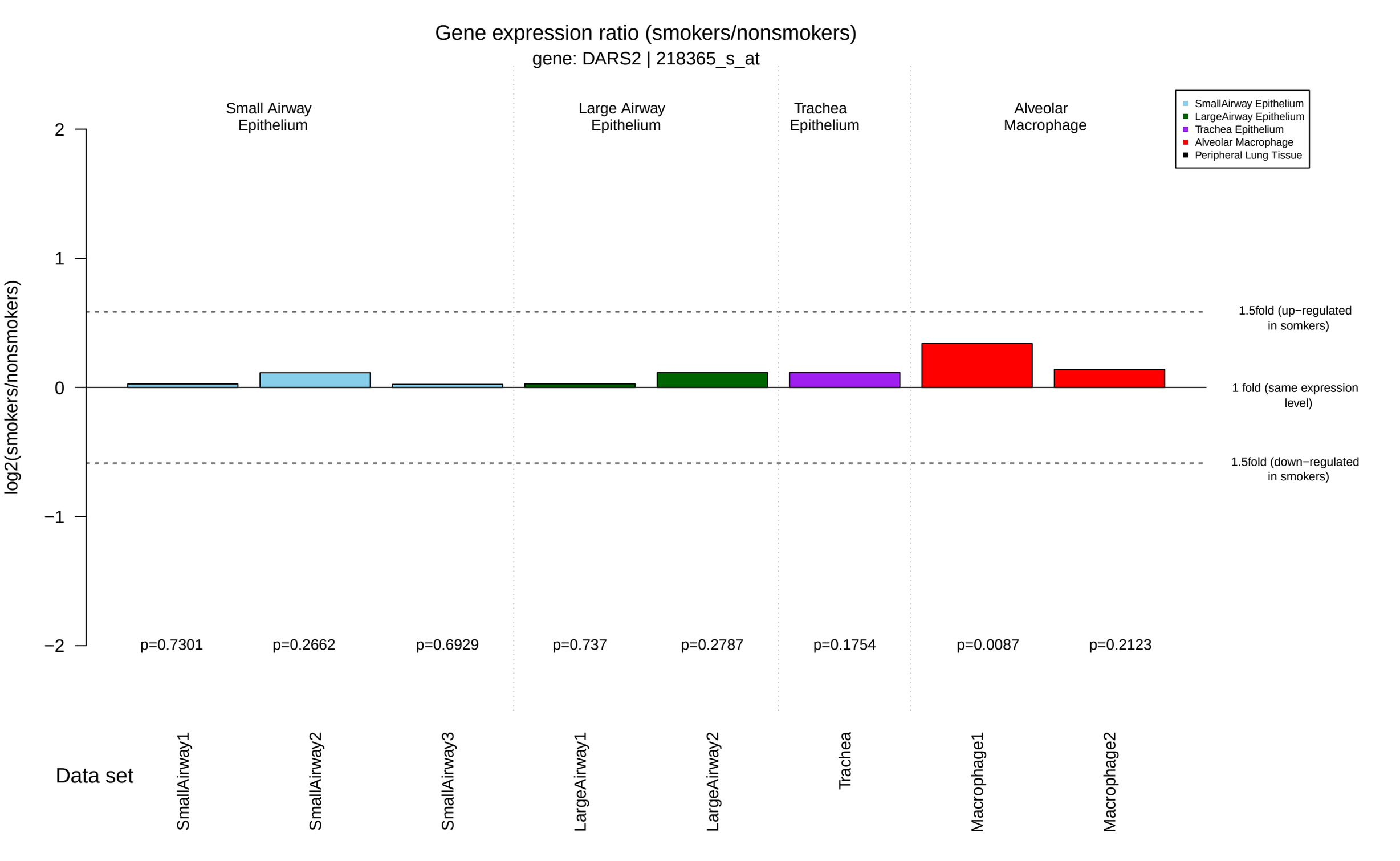
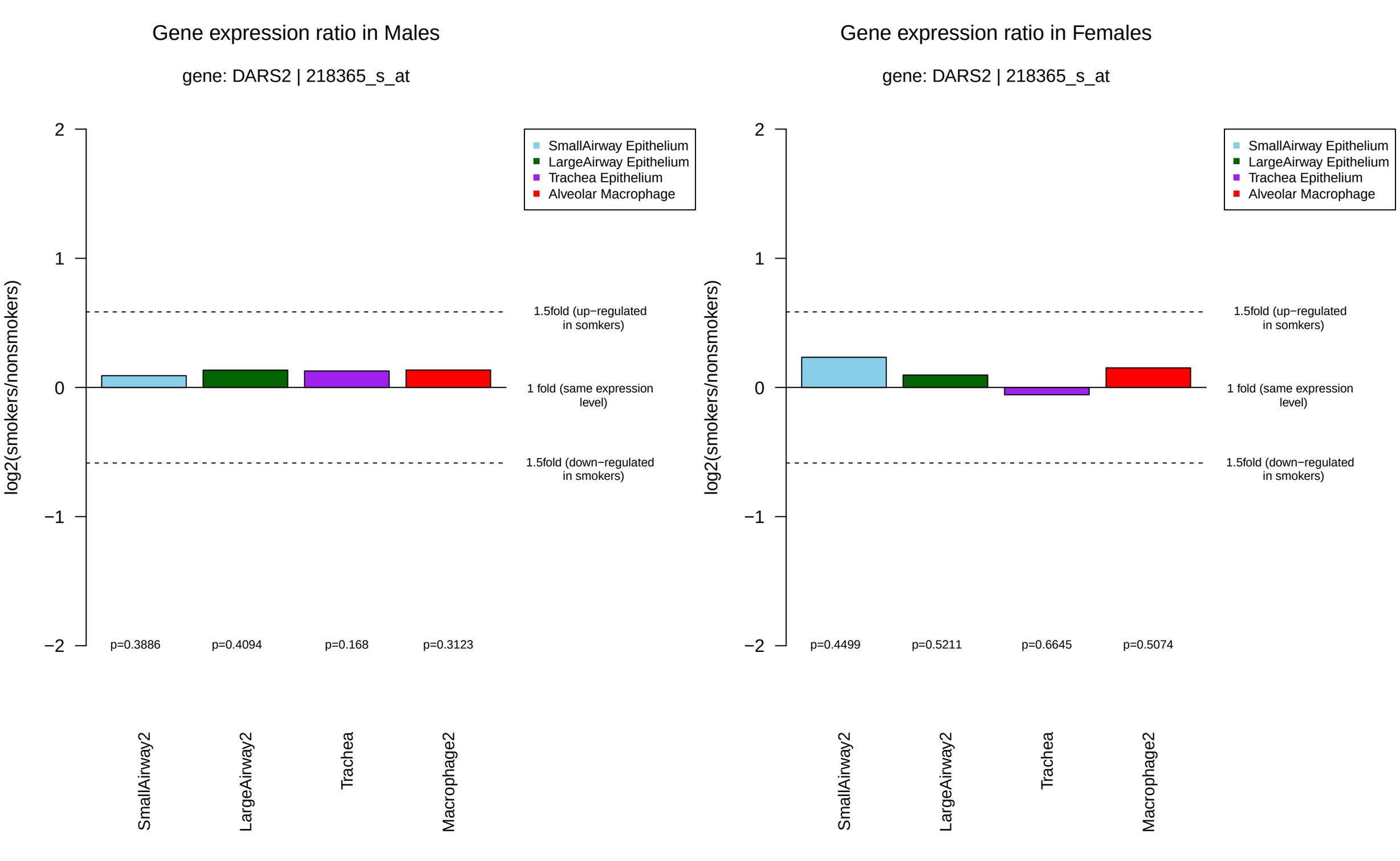
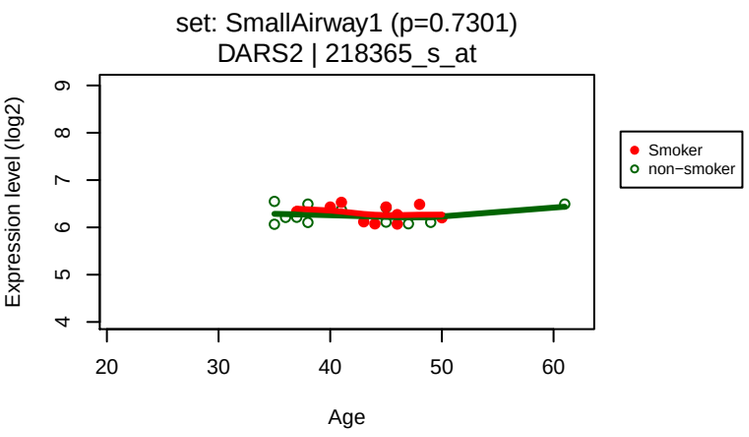
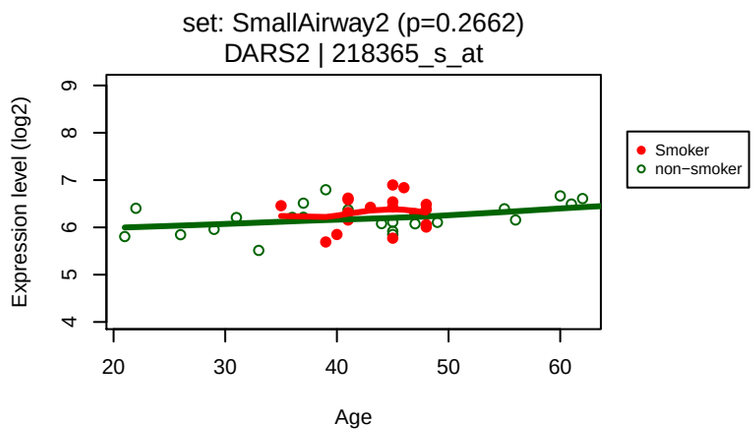
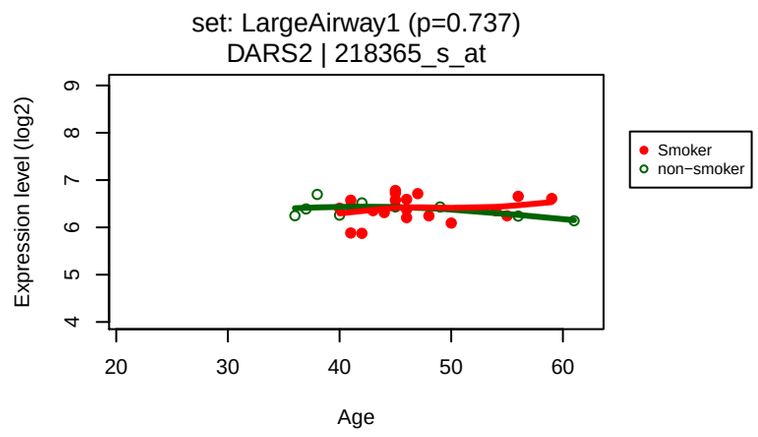

Gene expression of smokers and nonsmokers
gene: DARS2 | 218365_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7301
p=0.2662
p=0.6929
p=0.737
p=0.2787
p=0.1754
p=0.0087
p=0.2123
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

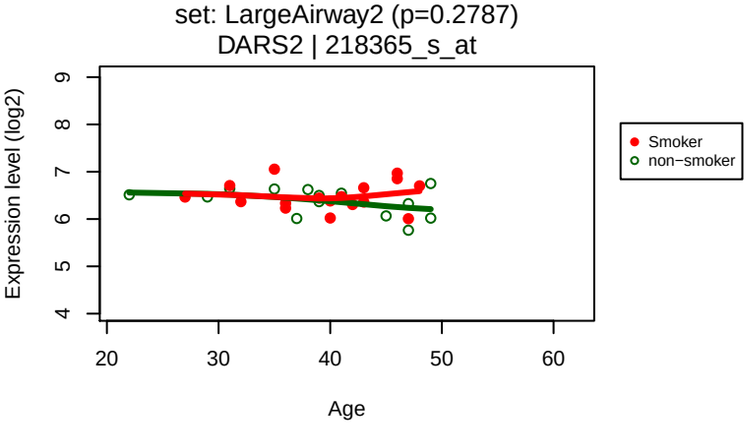

Gene expression ratio (smokers/nonsmokers)
gene: DARS2 | 218365_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7301
p=0.2662
p=0.6929
p=0.737
p=0.2787
p=0.1754
p=0.0087
p=0.2123
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

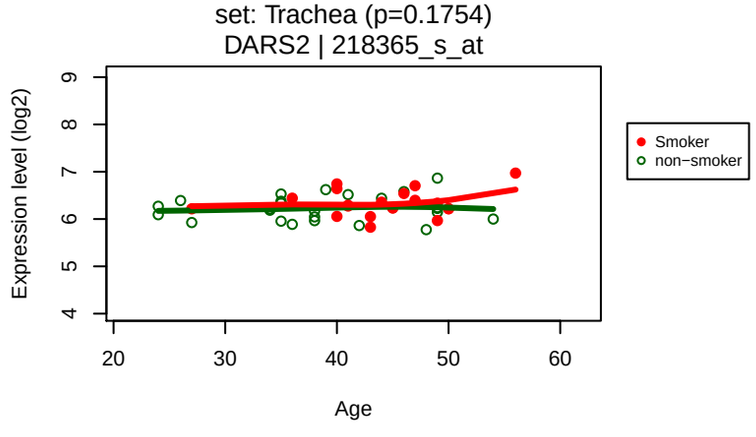

Gene expression ratio in Males
gene: DARS2 | 218365_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3886
p=0.4094
p=0.168
p=0.3123
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DARS2 | 218365_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4499
p=0.5211
p=0.6645
p=0.5074
SmallAirway2
LargeAirway2
Trachea
Macrophage2

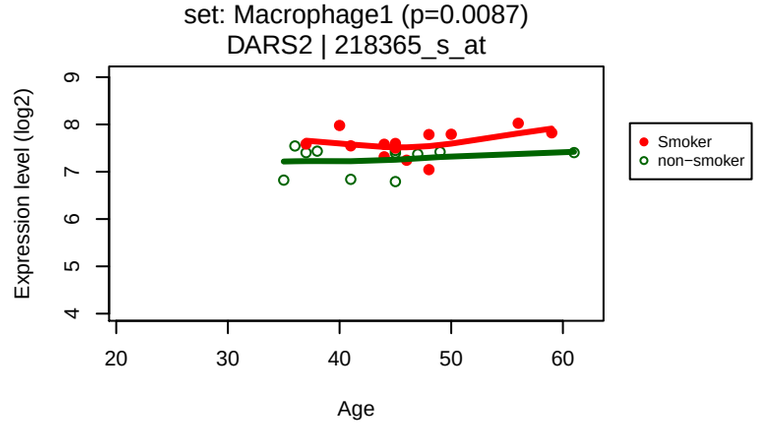

set: SmallAirway1 (p=0.7301)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker

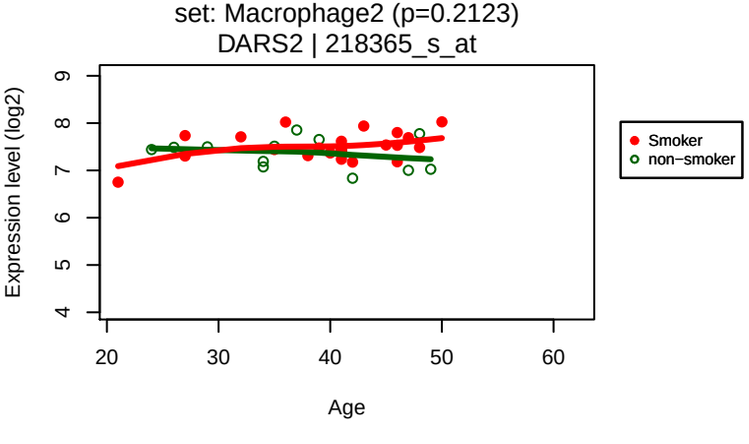

set: SmallAirway2 (p=0.2662)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker

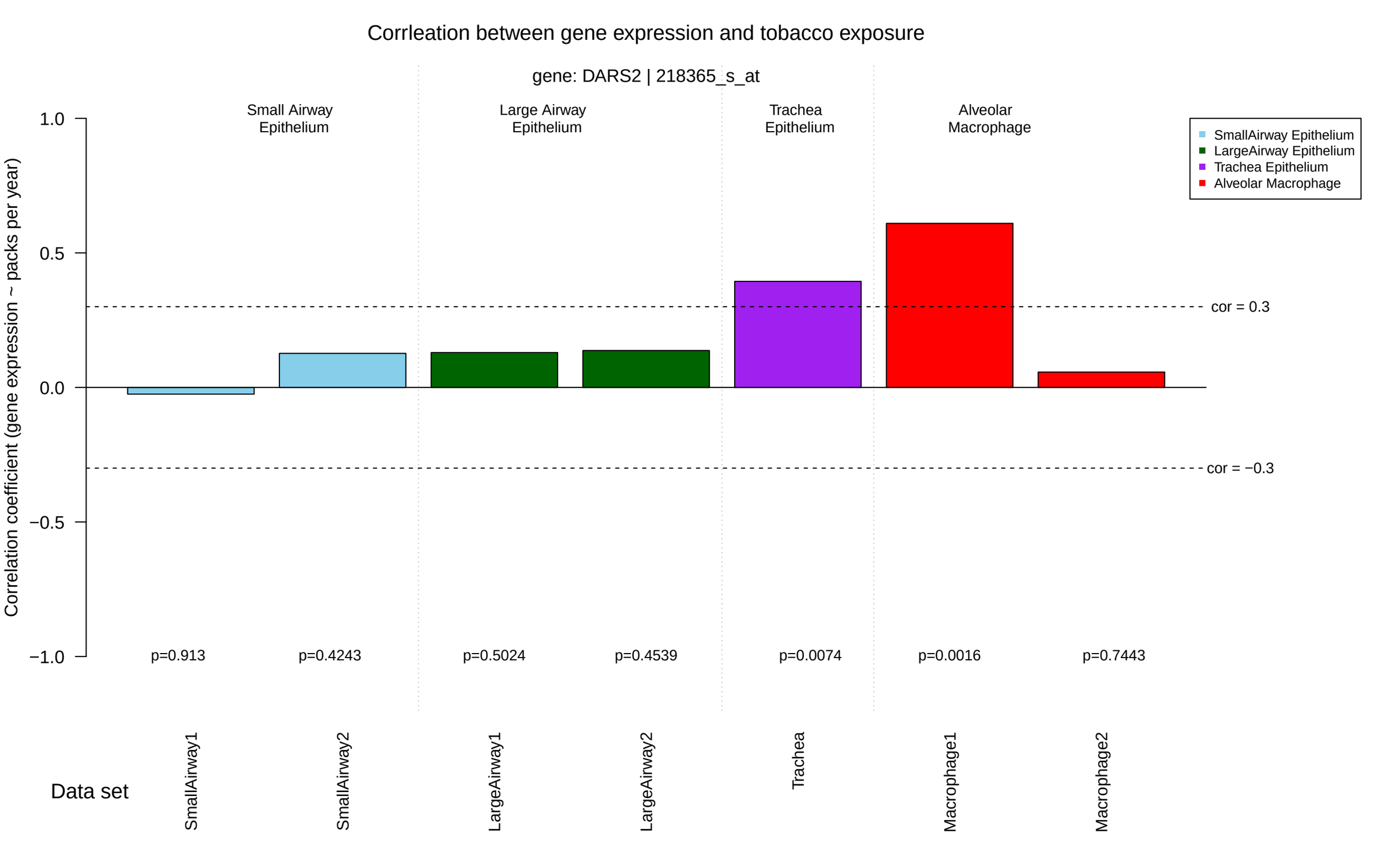

set: LargeAirway1 (p=0.737)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2787)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1754)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0087)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2123)
DARS2 | 218365_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DARS2 | 218365_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.913
p=0.4243
p=0.5024
p=0.4539
p=0.0074
p=0.0016
p=0.7443
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2